

1. Use mathematical induction to prove:

$$\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1}, \quad n \geq 1$$

2. What does the following function return? Can you prove that by mathematical induction?

```
MysteryFunction (positive integer n)
```

```
    i = 1  
    j = 5  
    while i ≠ n do  
        j = 2j - 3  
        i = i + 1  
    end while
```

```
return j
```

3. Use mathematical induction to show that `MysteryFunction(n)` returns $9n + 47 \cdot 4^{n-1} + 23$.

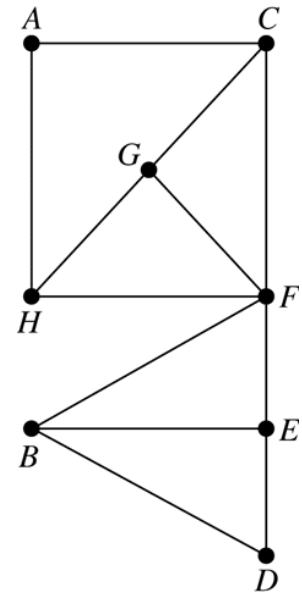
```
MysteryFunction (positive integer n)
```

```
    i = 1
    j = 79
    while i ≠ n do
        i = i + 1
        j = 4j - 27i - 33
    end while
```

```
return j
```

Graphs and Trees

1. Let $G = (V, E)$ be graph in the figure. How many paths are there from A to D? How many of these paths have length 6?



2. Consider a simple¹ graph $G = (V, E)$ with 30 edges.
 - a) What is the maximum number of vertices that G can have?
 - b) If G is connected, what is the maximum number of vertices that G can have?
 - c) What is the minimum number of vertices that G can have?

¹ A simple graph is an unweighted, undirected graph containing no loops or multiple edges between two vertices (do not confuse “loop” with “cycle”).

6. If a tree has three vertices of degree 2, four vertices of degree 3, four vertices of degree 4, and two vertices of degree 5, how many vertices with degree 1 does it have?
7. Let $T = (V, E)$ be a tree with 10 vertices. How many distinct paths are there in T ?

If all of the problems can't be done in the lab period, students can work on them on their own.